

BYRON JOSUE MARTINEZ

Data Engineer & AI Specialist | ETL Pipelines · Agentic Systems · On-Chain Analytics

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PROFESSIONAL SUMMARY

Data Engineer and AI Specialist with 5+ years building data-intensive systems end-to-end: from raw blockchain extraction and ETL pipeline engineering to agentic AI orchestration. I design and operate complete systems on my own hardware — no cloud vendor lock-in, optimized for cost and performance. Currently operating a Bitcoin ETL pipeline processing 948K+ blocks into ClickHouse OLAP and a 6-worker LangGraph agent system processing natural language tasks with local GPU inference — all at ~\$0.50/month.

CORE COMPETENCIES

- **ETL & Data Engineering:** Pipeline architecture, incremental loads, schema evolution, data quality checks, Parquet (zstd), ClickHouse File Engine, zero-duplication design
- **Agentic AI Systems:** LangGraph state machines, multi-agent orchestration, local LLM deployment (Ollama, Qwen 3 14B GPU), MCP tool servers, structured JSON outputs
- **Backend & APIs:** FastAPI, SSE streaming, REST design, Swagger/OpenAPI, Docker sandboxing, prompt injection defenses
- **Databases:** PostgreSQL (multi-table agent memory schemas), MySQL, SQLite, ClickHouse OLAP, query optimization
- **Blockchain Data:** Bitcoin Core RPC, UTXO normalization, block-level metrics, Binance API, reorg detection
- **Quantitative Analysis:** On-chain metrics, fee dynamics, UTXO cohort analysis, heavy-tail distributions, backtesting

TECHNICAL SKILLS

Python · Pandas · NumPy · Polars · FastAPI · LangGraph · LangChain · PostgreSQL · MySQL · SQLite · ClickHouse · Redis · Docker · Playwright · Git · Linux · Bitcoin Core RPC · Parquet · Jupyter · Power BI (DAX) · Qwen 3 14B · DeepSeek API · Ollama · MCP Servers · Web Scraping · Automatización · Prompt Engineering · Data Quality · Schema Evolution

PROFESSIONAL EXPERIENCE

Data Engineer & AI Specialist — Independent Projects | 2024 – Present

- Architected a 4-layer Bitcoin ETL pipeline processing 948K+ blocks with zero data duplication using Parquet (zstd) and ClickHouse File Engine
- Built Cowork-Local: a LangGraph multi-agent system with DeepSeek Pro planner, 5 specialized workers, PostgreSQL memory, SSE streaming, 90 Python files
- Deployed local LLM inference (Qwen 3 14B on NVIDIA RTX 4060 Ti) with 12 MCP tool servers at ~\$0.50/month operational cost
- Implemented state-file checkpointing, pause/resume/rollback, exponential backoff, and reorg detection for blockchain pipeline resilience

Operations Analyst — SGS Ecuador | 2022 – 2025

- Engineered resource allocation models reducing project delivery times by 15% across multi-site portfolio
- Deployed Power BI dashboards (DAX) for real-time operational tracking, replacing manual Excel workflows
- Audited and optimized certification workflows under ISO 9001, 14001, 45001, and 27001 standards

Financial Analyst — Banco Pichincha | 2019 – 2021

- Recalibrated credit risk models via feature engineering, improving prediction accuracy by 20% and reducing credit losses by 10%
- Automated data extraction and transformation workflows using Python and SQL, reducing manual reporting time
- Delivered quantitative risk exposure reports to senior management, directly informing lending policy adjustments

EDUCATION

Master's in Data Management and Technological Innovation — OBS Business School | 2021 – 2022

Bachelor's in Finance — Universidad San Francisco de Quito | 2011 – 2015

LANGUAGES

Spanish (Native) · English (Full Professional Proficiency — daily use in technical documentation & API integration)

CERTIFICATIONS

- Data Science & Machine Learning — Complete ML in Python, TensorFlow, Python & SQL for Data Science
- Analytics & BI — Excel & Power BI for Data Analyst (DAX, Power Query, dashboard design)
- ISO Internal Auditor — 9001, 14001, 45001, 27001:2022, 37001:2016, 22000-FSSC V6